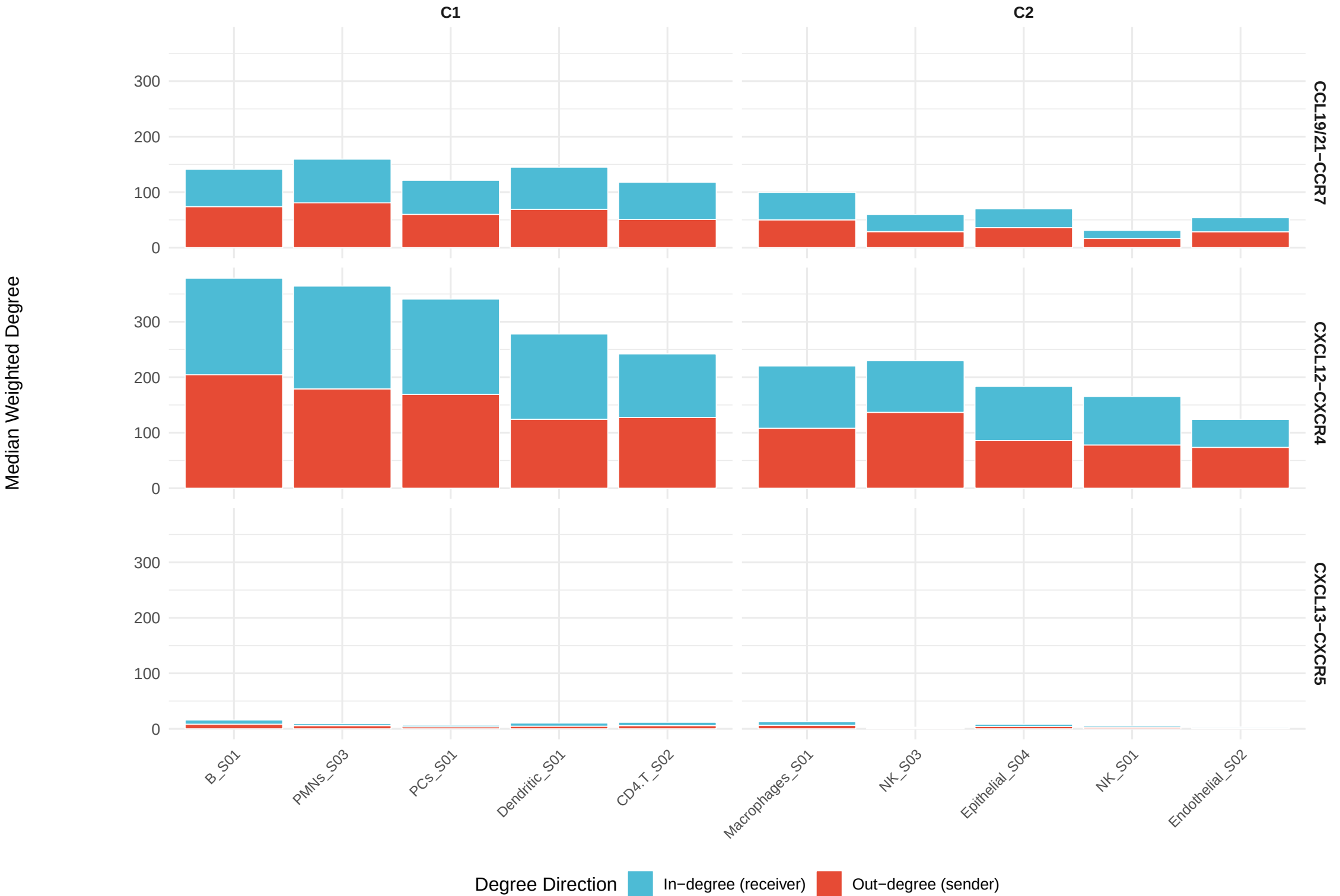


Directionality Analysis of Hub Cell State Network Interactions

Hub Cell State Degree Directionality

Weighted degree decomposed into sender (ligand) and receiver (receptor) components



Hub Cell State Network Role

Out-degree proportion (mean +/- SD across 3 chemokine axes)

